Organisational Values, Decision Fatigue and Inclusion: Evidence from a Field Experiment

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A Pre-analysis plan and pre-specified results

We examine the effect of the three treatment arms using the following regression:

$$\text{WomanWins}_{jcp} = \alpha + \beta_1 \text{Info}_j + \beta_2 \text{Comm}_j + \beta_3 \text{Info}_j \cdot \text{Comm}_j + \mu_p + \varepsilon_{jcp} \tag{A.1}$$

In this specification, WomanWins_{jcp} is an indicator equal to one if judge j on committee c votes for the woman in pair p. Info_j equals one if judge j receives the information treatment (treatments 2 and 4), and Comm_j equals one if judge j sits on a committee (treatments 3 and 4). The term μ_p denotes pair fixed effects and ε_{jp} is the error term. We cluster standard errors at the judge level.

For our secondary hypothesis we do subgroup analysis by gender. We look at the treatment effect for male and female judges separately using the specification in equation A.1. We run regression A.1 using OLS on the sub-sample of male and female judges and then compare the coefficients testing the same set of hypotheses for the comparison of the two sets of parameters.

We additionally estimate A.2 to study the effect of the triplet's gender composition.

$$\text{WomanWins}_{jcp} = \alpha + \beta_1 \text{Info}_j + \beta_2 \text{Woman}_c + \beta_3 \text{Info}_j \cdot \text{Woman}_c + \mu_p + \varepsilon_{jp}, \tag{A.2}$$

where $\text{Woman}_c = 1$ if there is a woman on the committee.

The results from these regressions are presented in Table A.1.

A.1 The effect of the treatments on the probability of dissent

Finally, we examine the outcomes at a committee level, examining the probability of a unanimous decision for each of the treatment arms. We first run the same regression as in equation A.1, but now at a committee level with as dependent variable whether or not the committee's decisions was unanimous.

We then run the same regressions with as outcome variables whether or not the committee's decision was unanimously for a male candidate, and whether or not the committee's decision was unanimously for a female candidate. These regressions help us examine the channels through which the treatments affect

	All judges b/se	Male judges b/se	Female judges b/se	Composition b/se
Information	0.038	0.043	0.007	-0.056
	(0.03)	(0.03)	(0.06)	(0.05)
Peer-Justification	0.014	0.025	-0.034	
	(0.03)	(0.03)	(0.06)	
Information × Peer-Justification	-0.045	-0.054	0.019	
	(0.04)	(0.04)	(0.08)	
Committee has at least one female member=1				-0.059
				(0.05)
Information × Committee has at least one female member=1				0.089
				(0.07)
Pair FE	Yes	Yes	Yes	Yes
Control mean	0.467	0.468	0.468	0.483
N	2858	2127	702	1513

Notes: This table implements Equation A.1 and A.2. Standard errors are clustered at the judge level and the regression includes pair fixed effects. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table A.1: The effects on the probability of a female candidate winning

judges' decisions. (For example, a high value for the committee treatment in the regression with as dependent variable UnanimousWomanWins would indicate judges in particular do not want to be found to be the only one dissenting by voting for the male candidate when the other judges' are expected to vote for the female candidate.)

We run the following regressions, varying the outcome as Unanimous, Unanimous Woman Wins, and Unanimous Man Wins:

$$\text{Unanimous}_{cp} = \alpha + \beta_1 \text{Info}_c + \beta_2 \text{Comm}_c + \beta_3 \text{Info}_c \cdot \text{Comm}_c + \mu_p + \varepsilon_{cp} \tag{A.3}$$

A.2 The effect of committee composition on the probability of dissent

Finally, for the subsample of judges on committees, we test whether committee composition affects the probability of dissent at a committee level. To do this, we use the same dependent variables as in regression A.2, but now use as a dependent variable the probability of a unanimous decision.

$$\text{Unanimous}_{cp} = \alpha + \beta_1 \text{Info}_c + \beta_2 \text{Woman}_c + \beta_3 \text{Info}_c \cdot \text{Woman}_c + \mu_p + \varepsilon_{cp} \tag{A.4}$$

	Unanimous b/se	Unan. Female b/se	Unan. Male b/se
Information	0.057	0.062*	-0.004
	(0.05)	(0.04)	(0.04)
Peer-Justification	0.081*	0.060*	0.021
	(0.05)	(0.03)	(0.04)
Information × Peer-Justification	-0.078	-0.082*	0.003
	(0.07)	(0.05)	(0.05)
Pair FE	Yes	Yes	Yes
Control mean	0.317	0.129	0.188
N	871	871	871

Notes This table implements Equation A.3. It shows the effect of the different treatment conditions on the probability of unanimity among the grouped triplets of judges who – within each treatment arm – assessed the same pair of candidates. We show effects (i) on unanimity, (ii) on unanimity in favour of the male candidate, and (iii) on unanimity in favour of the female candidate. Standard errors are clustered at the committee level and the regression includes pair fixed effects. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table A.2: The effect of the treatments on unanimity

	Unanimous b/se	Unan. Female b/se	Unan. Male b/se
Information	-0.009	-0.020	0.011
	(0.09)	(0.06)	(0.08)
Committee has at least one female member=1	0.137*	0.073	0.064
	(0.08)	(0.06)	(0.06)
Information × Committee has at least one female member=1	-0.028	0.019	-0.046
	(0.13)	(0.09)	(0.10)
Pair FE	Yes	Yes	Yes
Control mean	0.409	0.197	0.212
N	444	444	444

Notes This table implements Equation A.4. It shows the effect of the information treatment and gender composition of the triplet on the probability of unanimity among the grouped triplets of judges who – within each treatment arm – assessed the same pair of candidates. We show effects (i) on unanimity, (ii) on unanimity in favour of the male candidate, and (iii) on unanimity in favour of the female candidate. Standard errors are clustered at the committee level and the regression includes pair fixed effects. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table A.3: The effect of the Information and triplet composition on unanimity

B Model

In this appendix section, we present a stylised theoretical framework to capture the basic trade-offs that organisations face when they design assessment processes. On the one hand, the organisation wants decision-makers to use their discretion and good judgement in assessing candidates; on the other hand, the organisation cares about some dimensions in particular, and so does not want decision-makers to rely purely on their own personal and idiosyncratic preferences. This tension is a long-standing theme in organisational economics (for example, Dessein, 2002; Prendergast, 1993).

To capture this tension in our setting – and to consider potential organisational design options – we imagine different judges assessing candidates. For simplicity, imagine two judges (i and j). Each judge ℓ is characterised by a preferred evaluation rule $\omega_\ell \in [0, \pi/2]$, and chooses an evaluation rule $\theta_\ell \in [0, \pi/2]$.

Evaluators face three forces. First, they prefer to align their evaluation with their own view of what matters. Second, they face organisational pressure to emphasize particular dimensions. Third, they incur a cost from disagreeing with other evaluators (Bursztyn, Egorov, Haaland, Rao, & Roth, 2023). We capture these forces with the expected loss function:

$$L(\theta_i; \omega_i) = \underbrace{\frac{\alpha}{2}\left(\theta_i - \omega_i\right)^2}_{\text{personal preference}} + \underbrace{\beta\left(\frac{\pi}{2} - \theta_i\right)}_{\text{organizational alignment}} + \underbrace{\gamma\, \mathbb{E}_{\omega_j}[\,|\theta_i - \theta(\omega_j)|\,]}_{\text{dissent aversion}}. \tag{B.5}$$

where $\alpha > 0$ governs the cost of deviating from one's own preferred evaluation, β captures organizational pressure toward the organizational benchmark $\pi/2$, and $\gamma > 0$ captures aversion to disagreement.

B.1 Probability of disagreement and geometric intuition

To motivate the disagreement term, it is helpful to consider a geometric interpretation. Suppose candidates differ along two dimensions, (x, y), where y captures the dimension valued by the organisation and x captures an idiosyncratic or stylistic trait. For simplicity, assume (x_k, y_k) are drawn independently from a bivariate normal distribution with mean zero and equal variance across dimensions. The key property we rely on is the radial symmetry of the joint distribution.

Each judge i chooses an evaluation angle $\theta_\ell \in [0, \pi/2]$, which defines an evaluation vector $\mathbf{v}_\ell = (\cos\theta_\ell, \sin\theta_\ell)$. The score assigned to candidate k by judge ℓ is $s_\ell(k) = \cos(\theta_\ell) \cdot x_k + \sin(\theta_\ell) \cdot y_k$, and the judge votes for candidate 1 if and only if $s_\ell(1) \geq s_\ell(2)$.

Under symmetry, disagreement between two evaluators arises when their decision rules rank candidates differently. In this setting, the probability of disagreement is proportional to the angular distance between decision rules, as formalised in the following lemma.

Lemma 1. *For any evaluation angles θ_i, θ_j,*

$$\Pr(\text{disagree}) \;=\; \frac{|\theta_i - \theta_j|}{\pi}.$$

Proof. Judge ℓ votes for candidate 1 iff $X_\ell := \cos(\theta_\ell)\, D_x + \sin(\theta_\ell)\, D_y > 0$, where $D_x = x_1 - x_2$ and $D_y = y_1 - y_2$ are independent $\mathcal{N}(0, 2)$ random variables. The pair (X_i, X_j) is bivariate normal with zero means and correlation $\rho = \cos(\theta_i - \theta_j)$. The radial symmetry of (D_x, D_y) implies that the disagreement region consists of two opposing wedges of opening angle $|\theta_i - \theta_j|$. ∎

B.2 Evaluator problem and first-order condition

Evaluator i chooses θ_i to minimise expected loss as defined above. Assuming an interior solution and differentiability almost everywhere, the first-order condition is:

$$\alpha(\theta_i - \omega_i) - \beta + \gamma\, \mathbb{E}_{\omega_j}\left[\text{sgn}(\theta_i - \theta(\omega_j))\right] = 0. \tag{B.6}$$

We look for a symmetric equilibrium in which evaluators adopt a monotone strategy $\theta(\omega)$. Depending on parameter values, corner solutions or bunching at the boundary may arise. We focus initially on the interior linear equilibrium for expositional clarity. (We also focus on parameter values such that $B > 0$, ensuring that the equilibrium is monotone in type. When $B < 0$, the equilibrium is non-monotone and unintuitive to interpret, so we do not consider this case.)

Under monotonicity of $\theta(\cdot)$, we have

$$\text{sgn}(\theta_i - \theta(\omega_j)) = \text{sgn}(\omega_i - \omega_j),$$

so the expectation depends only on the distribution of ω_j. This reflects the fact that, under monotone strategies, disagreement depends only on the relative ranking of types.

With $\omega \sim U[0, \pi/2]$, this expectation simplifies to:

$$\mathbb{E}_{\omega_j}\left[\text{sgn}(\omega_i - \omega_j)\right] = \frac{4}{\pi}\left(\omega_i - \frac{\pi}{4}\right). \tag{B.7}$$

B.3 Equilibrium

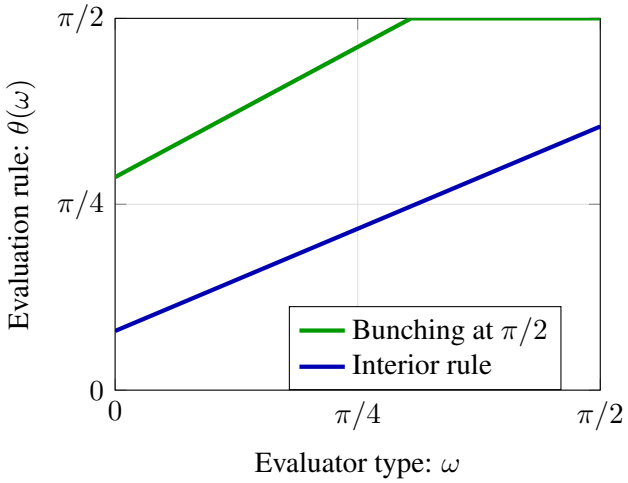
In equilibrium, evaluators trade off these forces when choosing their evaluation rule. We focus on symmetric equilibria in which strategies are monotone in type. The following proposition characterises the equilibrium.

Proposition 1. *There exists a symmetric equilibrium in which evaluators adopt a linear decision rule*

$$\theta(\omega) = A + B\omega,$$

for interior types, with possible bunching at the boundary. The coefficients are given by

$$B = 1 - \frac{4\gamma}{\pi\alpha}, \quad A = \frac{\beta}{\alpha} + (1 - B)\frac{\pi}{4}.$$



Note: The figure illustrates the equilibrium mapping $\theta(\omega)$. The interior rule is linear. When organisational pressure is sufficiently strong, the rule is truncated at the organisational benchmark $\pi/2$, generating bunching and a high-alignment outcome. In this region, further increases in β do not affect behaviour.

Figure A.1: Equilibrium evaluation rules

The linear rule above describes interior behaviour. However, the choice set is bounded: $\theta \in [0, \pi/2]$. For sufficiently high values of β relative to α and γ, the linear rule may imply $\theta(\omega) \geq \pi/2$ for a range of types. In this case, evaluators bunch at the boundary $\theta = \pi/2$, fully implementing the organisational criterion. Formally, the equilibrium takes the truncated form

$$\theta(\omega) = \min\left\{\frac{\pi}{2},\, A + B\omega\right\}.$$

This mapping is illustrated in Figure A.1.

B.4 Proof of Proposition 1

Substituting into the first-order condition:

$$\alpha(\theta_i - \omega_i) - \beta + \gamma \cdot \frac{4}{\pi}\left(\omega_i - \frac{\pi}{4}\right) = 0. \tag{B.8}$$

Rearranging:

$$\theta_i = \omega_i + \frac{\beta}{\alpha} - \frac{4\gamma}{\pi\alpha}\left(\omega_i - \frac{\pi}{4}\right). \tag{B.9}$$

Collecting terms:

$$\theta_i = \frac{\beta}{\alpha} + \frac{4\gamma}{\pi\alpha} \cdot \frac{\pi}{4} + \left(1 - \frac{4\gamma}{\pi\alpha}\right)\omega_i. \tag{B.10}$$

Simplifying:

$$\theta_i = A + B\omega_i, \tag{B.11}$$

where

$$B = 1 - \frac{4\gamma}{\pi\alpha}, \quad A = \frac{\beta}{\alpha} + (1 - B)\frac{\pi}{4}. \tag{B.12}$$

This establishes the linear equilibrium stated in Proposition 1.

B.5 Interpretation

The interior equilibrium can be interpreted as a weighted average. Evaluators choose a rule that combines their own preferred weighting ω_i with a common benchmark:

$$\theta(\omega_i) = \frac{\beta}{\alpha} + (1 - B)\frac{\pi}{4} + B\omega_i,$$

so that B governs the weight placed on individual preferences ω_i, while $1 - B$ captures the extent of convergence toward a shared benchmark (centered at $\pi/4$).

The parameters α, β, and γ map directly into this structure. Higher α increases reliance on individual preferences (raising B), higher β shifts evaluations toward the organisational benchmark (raising A), and higher γ compresses variation in evaluations by reducing B and inducing convergence across evaluators. Notably, β operates as a pure level shift, while both α and γ also affect dispersion and therefore indirectly influence the level of evaluations. These forces map closely to the experimental design: fatigue can be interpreted as increasing the effective weight on evaluators' own preferred criteria, which in the model corresponds to a higher effective α; organisational messages affect β; and features that increase the salience or cost of disagreement between evaluators affect γ.

The model can be interpreted as implying differing levels of alignment across evaluators. When organisational pressure is sufficiently strong relative to individual preferences, evaluators bunch at the boundary $\theta = \pi/2$, generating a high-alignment (pooling-like) outcome with low dispersion. In such cases, organisational criteria are already effectively implemented. This provides a natural interpretation of why the Information treatment has little effect in early rounds of the experiment. Over time, decision fatigue can be interpreted as increasing the relative cost of deviating from the personal bliss point: α increases, judges rely more on their own preferences, and alignment declines. But an increase in organisational alignment – for example, through clearer emphasis to judges on organisational objectives – can generate a countervailing increase in β, sustaining a more aligned equilibrium. Heterogeneity in responses arises naturally in intermediate cases where some evaluators remain close to the organisational benchmark while others do not. In addition, the effect of disagreement aversion depends on the level of organisational alignment: when alignment is high, social pressure reinforces organisational objectives, while when alignment is weaker, judges may instead converge toward a common rule, reflecting social pressure to agree.

C Details experiment

C.1 Protocol for the control group

This subsection details the protocol for the business plan competition for those judges not assigned to a treatment, (the control group). Following this we then detail how the protocol is different for those receiving the Peer-Justification treatment. There are no further details to report on the Information treatment.

Judges are invited in groups of 16 to a hotel in central Addis Ababa. Once the judges arrive at the hotel and register, the assessments will start at a fixed time after which no new judges are included in the main sample. To start, the judges enter a room set up in a classroom style format and watch two videos. The first video explains the full format of the assessments and how the winner will be decided; the second video features a prominent Ethiopian businessman discussing the importance of access to start-up capital for aspiring entrepreneurs in Ethiopia (and, thus, the importance of the business plan competition). After watching these two videos, and after having had an opportunity to ask questions after the first video, the responses to which are strictly pre-defined, the assessments start.

The respondents then start the assessments in which they will watch the three-minute recordings of twelve pairs of candidates in the business plan competition. After watching each pair, which consists of one female and one male candidate, they are asked which of these candidates they want to cast their vote for for the business plan competition and how difficult it was to make this decision. The respondent answers these questions without showing the enumerator their answer.

These assessments provide the data for the main experiment, with as primary outcome whether the judge votes for the female candidate. Following the assessments, we then return to one of the pair of candidates and ask the judge to provide feedback. The enumerator observes, using the survey software, who the judge voted for for one pair of candidates. The respondent watches the recordings of these two candidates again, and is asked to write down reasons for their decision and to provide some feedback for the candidates both on some fixed dimensions and open-ended feedback. Finally, the judge is asked the following question for both a male and female judge:

> Please now imagine a [male/female] judge with [x] years of managerial experience working as [position] in [industry] has also assessed these candidates. We would like to ask you whom you think they would have voted for.

The purpose of this question is to elicit beliefs on how other judges would vote. The specific set-up is designed to mirror the committee treatment. These fields are filled based on the set of other judges who are invited to attend the same assessment session, i.e. if judges A and B both attend the same assessment session, judge A may here see these characteristics for judge B. Note that while the position, industry and years of experience are based on the actual characteristics, the gender may be changed to ensure the judge is asked about both a male and female judge.

We then run a "feedback session" with the individual judge in which they first, privately, write down reasons for their decision. They then give feedback on a number of fixed dimensions taken from

Fafchamps and Quinn (2018) before being given the opportunity to provide some open-form feedback to the candidate. This feedback will then, after the conclusion of the full experiment, be shared with the candidate.

To finish the assessment day, judges are asked a number of closed form-questions about the experiment, about their relationships with other judges and on the ambivalent sexism inventory Glick and Fiske (1997); see the separately attached document 'post-competition questions' for the full questionnaire. We implement this after the experiment in order to avoid the possibility of experimenter demand effects.

The entire experiment is conducted in Amharic.

C.2 Further details on the Peer-Justification Treatment

The protocol for judges assigned to the Peer-Justification treatment differs from that for individual judges as follows, and is explained in a separate explanatory video to that for individual judges:

- The room will be set up boardroom rather than classroom style so judges can see each other, and crucially the judges with whom they are on a committee.
- At the start of the assessment day, judges are asked to introduce themselves by telling other judges their name, industry and their position at their company. The judges are asked not to share any additional information.
- Judges still go through the decisions individually, but know they are deciding together with two other judges in the room. Before they watch the videos for each pair of candidates, they are shown photo CV's of these two judges. This includes, beyond a photo of the judge, their name, industry and position at their company. They are also reminded their decisions may be shared with these judges in the feedback sessions.
- The judges are on a different random triplet of judges for each pairwise assessment.
- After the twelve assessments, they also re-watch one of the pairs of candidates' responses. They are then asked who they think *the other two judges on this triplet* voted for. They again see the exact same set of characteristics as those the individual judges see based on the photo CV, and now additionally know this judges' name and what they look like.
- The three judges then come together and are told who each of them voted for for the pair of candidates whose videos they just re-watched. They are asked to each give reasons for their decisions to each other as the judges not receiving the Peer-Justification treatment judges did individually. They then, together, provide the same feedback as an individual judge to the candidate.
- The judges know that in the case of a split decision, the candidates will get respectively 2 and 1 point, *i.e.* one point for each vote that is cast for them.

To ensure judges on the Peer-Justification treatment are aware of the section at the end of the experiment in which their decisions are made public to other judges. To do so, the feedback session is clearly highlighted in the video at the start of the assessments explaining the protocol. In this video, actors playing judges go through the full experiment and are shown discussing their decisions (with the sound muted; the respondents know these are actors). Beyond this, judges are reminded every time they are shown the judges on their triplet that they may have to give reasons for their decisions to these two judges.

C.3 Randomization

This section details how randomisation is done for the experiment.

Assignment of judges to treatments Judges are first invited to participate as a judge in the business plan competition. We aim to over-sample judges due to high expected non-attendance, thus intending to invite 320 judges in total aiming to have 240 actually attend including 180 male and 60 female judges. After inviting the judges, we randomly assign them to a treatment assigning 20 women and 60 men to each treatment. We thus assign 80 judges assigned to each treatment. Once a judge is assigned to one treatment, they will not be swapped across treatments. To specify how randomisation is done, once we have a sample of judges we will randomly split judges across the treatments using Stata's "cut" command based on a random uniform variable with no duplicates. Using this command, the set of male and female judges are separately assigned to the four treatments resulting in a quarter of the male and female judges in the sample being assigned to each treatment.

Order of sessions We will run five or six sessions for each treatment arm over the course of two weeks, running two sessions each day. Every two days, fours sessions, one for each of the four treatment arms, will be conducted during this time (except on Sunday). This ensures ex-ante attendance for each treatment arm is expected to be the same, and we impose the same constraints for inclusion on individual and committee judges.

Assignment of judges to sessions We flexibly assign judges to a specific assessment day. Judges are randomly assigned to and invited for a specific assessment day, but if they cannot attend this slot they are offered to attend one of the other four slots for their treatment. For each assessment day, we aim to include at least ten male and two female judges. To achieve this, we invite eleven male and four or five female judges to each assessment day. We will conduct the experiment if more than eight judges show up, if fewer than eight judges attend the assessments will be rescheduled. Note that to do the feedback sessions at the end of the sessions for the committee treatment, we require a multiple of three judges to attend for each judge can be included in a three-member assessment. To maximise the number of judges we can include, we allow for some judges not to do a feedback session for the committee treatment if they cannot do so as part of a three-member committee.

Assignment of judges to triplets The final element of randomisation is due to the difficulty in predicting how many judges will attend. On the day, each judges is randomly assigned a "judge number" from one to the number of judges attending. These judges numbers have been randomly pre-assigned to be members of a specific set of triplets assessing specific pair of candidates. This method allows us to flexibly deal with attrition in a logistically achievable way.

Assignment of candidates to competitions Finally, we randomly assign candidates to competitions. We first stratify by gender, to then randomly assign five male and five female candidates to each competition.

C.4 Summary statistics judges

Table A.4: Judge-level summary statistics

	All	Control	Information	Peer-Justification	Information & Peer-Justification	p-value
Gender (1=male)	0.741	0.702	0.793	0.722	0.750	0.702
Judge age	41.41	38.91	42.62	41.15	43.04	0.246
Experience in current position (years)	6.12	6.49	6.81	5.93	5.27	0.474
Total experience (years)	19.68	19.04	19.95	19.91	19.78	0.981
Formal management education	0.770	0.772	0.793	0.722	0.804	0.696
At least a BA	0.782	0.772	0.810	0.806	0.732	0.719
HR	0.407	0.491	0.345	0.403	0.393	0.452
Administration	0.342	0.281	0.431	0.306	0.357	0.327
Finance	0.128	0.140	0.103	0.125	0.143	0.918
Other	0.123	0.088	0.121	0.167	0.107	0.565
Most senior manager or owner	0.329	0.298	0.431	0.278	0.321	0.281
Finance and administration	0.148	0.158	0.155	0.125	0.161	0.933
HR Manager	0.325	0.386	0.259	0.333	0.321	0.545
Other	0.198	0.158	0.155	0.264	0.196	0.362

Notes This table displays the average characteristics of judges in each of the treatment arms. The department of the manager is the main department in which the manager operates in their firm, the other category contains all departments in which fewer than 5% of managers work. Columns two to six report these characteristics by treatment, and column seven reports the p-value for a Wald test for the hypothesis that these means are equal across the treatment arms. The table uses data for the 243 judges, out of 245, that filled out our short follow up survey after the assessments.

C.5 Summary statistics expert rankings

	Male candidates	Female candidates	Male minus female	p-value
Expert 1 Score	10.722	10.020	0.702	0.003
Expert 2 Score	14.833	15.204	-0.371	0.191
Average Score	12.778	12.612	0.165	0.457
Expert 1 choice	0.498	0.380	0.118	0.048
Expert 2 choice	0.400	0.518	-0.118	0.053
Expert favourite	0.461	0.473	-0.012	0.843
Unanimous expert favourite	0.286	0.261	0.024	0.605

Notes: This table describes the expert scores for male and female candidates, and tests whether the two HR experts assess male and female candidates to be similar on average. The table reports Expert 1 Score and Expert 2 Score (the average raw score from 1-20 for each of the experts), the average score (averaged by candidate over the two experts), the probability of being Expert 1 and Expert 2 favourite, the probability of being the "expert favourite", i.e. preferred by the expert based on the average score, and the "unanimous expert favourite, i.e. preferred by both experts.

Table A.5: Summary statistics of expert rankings

D Additional Results

	Woman Wins	Expert-favoured Woman Wins	Expert-favoured Man Wins
Round judge	-0.014**	-0.008**	0.004
	(0.006)	(0.004)	(0.004)
Information × Round judge	0.021***	0.016***	-0.005
	(0.008)	(0.005)	(0.006)
Peer-Justification × Round judge	0.011	0.009*	-0.002
	(0.007)	(0.005)	(0.005)
Information & Peer-Justification × Round judge	0.016**	0.011**	-0.003
	(0.008)	(0.005)	(0.006)
Information	-0.101*	-0.057	0.034
	(0.057)	(0.037)	(0.041)
Peer-Justification	-0.059	-0.045	0.004
	(0.055)	(0.036)	(0.040)
Information & Peer-Justification	-0.095*	-0.062	0.018
	(0.057)	(0.038)	(0.042)
Pair FE	Yes	Yes	Yes
Control mean (early rounds)	0.502	0.284	0.294
N	2,858	2,858	2,858

Notes: Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table A.6: The treatment effect interacted with rounds (continuous)

	Male Score	Female Score	Score Diff.	Share Fem. Fav.	N
1	12.727	12.505	-0.221	0.454	183
2	12.737	12.608	-0.129	0.479	190
3	12.727	12.608	-0.120	0.465	200
4	12.769	12.508	-0.261	0.447	199
5	12.650	12.612	-0.037	0.476	187
6	12.722	12.688	-0.034	0.485	194
7	12.716	12.682	-0.035	0.473	201
8	12.848	12.557	-0.291	0.443	194
p-value (joint)	0.999	0.996	0.990	0.989	.

Notes: Each row corresponds to one of eight treatment-arm $\times$ round-half cells: rows 1–4 cover the early rounds (1–6) of the control, Information, Peer-Justification, and Information & Peer-Justification treatment, respectively; rows 5–8 cover the late rounds (7–12) of the same four arms. "Male Score" and "Female Score" are the mean expert evaluation scores of the male and female candidates in each cell; "Score Diff." is the female-minus-male mean score; "Share Fem. Fav." is the fraction of pairs in which the female candidate has the higher average expert score; N is the number of distinct candidate pairs contributing to the cell. The sample is collapsed to the pair level within each cell so that each pair contributes once. The final row reports the p-value of a one-way ANOVA F-test of equality of the cell means across all eight cells. Statistical significance is denoted by * ($p < 0.10$), ** ($p < 0.05$), and *** ($p < 0.01$).

Table A.7: Balance of expert scores across treatment arms and rounds

	(1) Woman Wins	(2) Expert-favoured Woman Wins	(3) Expert-favoured Man Wins
Late Round	-0.099**	-0.068**	0.026
	(0.045)	(0.028)	(0.030)
Pair FE	Yes	Yes	Yes
Judge FE	Yes	Yes	Yes
Mean (early rounds)	0.500	0.284	0.293
N	651	651	651

Notes: This table restricts the sample to the control group and reports OLS estimates of the difference between late rounds (7–12) and early rounds (1–6). The dependent variables are indicators for whether the woman wins, whether the expert-favoured woman wins, and whether the expert-favoured man wins. All specifications include candidate-pair and judge fixed effects. Standard errors are clustered at the judge level and reported in parentheses. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table A.8: Fatigue in the control group with judge fixed effects

	(1) Woman Wins	(2) Expert-favoured Woman Wins	(3) Expert-favoured Man Wins
Late Round	-0.103**	-0.083***	0.025
	(0.042)	(0.027)	(0.029)
Information × Late Round	0.122**	0.116***	-0.011
	(0.053)	(0.036)	(0.036)
Peer-Justification × Late Round	0.070	0.072**	-0.004
	(0.051)	(0.034)	(0.035)
Information & Peer-Justification × Late Round	0.101*	0.083**	-0.016
	(0.055)	(0.037)	(0.037)
Pair FE	Yes	Yes	Yes
Judge FE	Yes	Yes	Yes
Control mean (early rounds)	0.502	0.284	0.294
p: Information differential late change	0.022	0.001	0.756
p: Peer-Justification differential late change	0.168	0.038	0.913
p: Information & Peer-Just. differential late change	0.066	0.025	0.659
N	2,858	2,858	2,858

Notes: This table reports OLS estimates from treatment-by-late-round specifications, where late rounds are rounds 7–12 and early rounds are rounds 1–6. The dependent variables are indicators for whether the woman wins, whether the expert-favoured woman wins, and whether the expert-favoured man wins. All specifications include candidate-pair and judge fixed effects; treatment main effects are absorbed by the judge fixed effects. Standard errors are clustered at the judge level and reported in parentheses. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table A.9: Treatment effects in late rounds with judge fixed effects

D.1 Session-level clustering

Our preferred specification clusters at the judge level, the unit of treatment assignment. As a robustness check, Tables A.10 and A.11 report wild-cluster bootstrap p-values using session-level Webb weights to allow for shared shocks within sessions. The central finding is unchanged: the probability that an expert-favoured woman wins declines in late rounds in the control group, and the Information treatment offsets this deterioration. Both the decline and the offsetting treatment interaction remain statistically significant. Inference for female selection overall is less precise, although the Information-by-late-round interaction remains significant.

	(1) Woman Wins	(2) Expert-favoured Woman Wins	(3) Expert-favoured Man Wins
Late Round	-0.099	-0.070**	0.027
	[0.106]	[0.024]	[0.503]
Mean (early rounds)	0.500	0.284	0.293
N	651	651	651
Sessions	5	5	5
Pair FE	Yes	Yes	Yes

Notes: This table reports OLS estimates of late-round effects on judge decisions using only the control group, regressing each outcome on a Late Round indicator (rounds 7–12). The dependent variables are: (i) woman wins, (ii) expert-favoured woman wins ($= 1$ if the female candidate is both expert-favoured and wins); and (iii) expert-favoured man wins ($= 1$ if the male candidate is both expert-favoured and wins). The omitted category is Early Rounds (rounds 1–6). All specifications include pair fixed effects. Wild-cluster bootstrap p-values using session-level weights are reported in square brackets; significance stars are based on these bootstrap p-values. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table A.10: Gender Bias in the Control Group

	(1) Woman Wins	(2) Expert-favoured Woman Wins	(3) Expert-favoured Man Wins
Late Round	-0.103*	-0.082**	0.026
	[0.077]	[0.023]	[0.374]
Information × Late Round	0.135**	0.120**	-0.021
	[0.032]	[0.019]	[0.517]
Peer-Justification × Late Round	0.068	0.071*	-0.004
	[0.232]	[0.058]	[0.895]
Information & Peer-Justification × Late Round	0.098	0.084**	-0.014
	[0.108]	[0.040]	[0.708]
Information	-0.030	-0.015	0.010
	[0.196]	[0.599]	[0.619]
Peer-Justification	-0.021	-0.024	-0.003
	[0.576]	[0.469]	[0.925]
Information & Peer-Justification	-0.043	-0.036	0.005
	[0.371]	[0.310]	[0.890]
Control mean (early rounds)	0.502	0.284	0.294
p: Information (late)	[0.132]	[0.036]	[0.784]
p: Peer-Justification (late)	[0.414]	[0.043]	[0.864]
p: Information & Peer-Justification (late)	[0.426]	[0.082]	[0.828]
N	2,858	2,858	2,858
Sessions	21	21	21
Pair FE	Yes	Yes	Yes

Notes: This table reports OLS estimates of treatment effects on individuals' decision outcomes, interacted with evaluation timing. The dependent variables are: (i) woman wins, (ii) expert-favoured woman wins ($= 1$ if the female candidate is both expert-favoured and wins); and (iii) expert-favoured man wins ($= 1$ if the male candidate is both expert-favoured and wins). Treatment indicators—Information, Peer-Justification, and Information & Peer-Justification—are each interacted with *Late round*, an indicator for rounds 7–12 (late rounds). The omitted category is Control × early rounds. p-values at the bottom test whether the total treatment effect in late rounds (main + interaction) equals zero. "Control mean (early rounds)" reports the mean of the dependent variable for the control group in rounds 1–6. All specifications include pair fixed effects. Wild-cluster bootstrap p-values using session-level weights are reported in square brackets; significance stars are based on these bootstrap p-values. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table A.11: Treatment effects in early versus late rounds

D.2 Sensitivity to expert favourite definition

Our analyses frequently condition on whether the female candidate is expert-favoured, defined as having a higher average expert score. Panel A of Table A.12 shows that the main treatment effects are robust to alternative definitions: a strict threshold requiring a score difference exceeding half a standard deviation, and a requirement that both individual experts agree on the female candidate's superiority.

Panel B of Table A.12 extends this by including the treatment $\times$ late-round interaction under each definition. The emergence of bias in late rounds and its prevention by the Information treatment are stable across expert-favourite operationalisations.

E External Validation: Gender Bias in Candidate Evaluation

In a separate experiment, HR managers from the same pool of firms evaluated young professionals for their suitability for entrepreneurship and a management position in their firm through video vignettes.[1] Three features of this experiment make it an ideal setting for our purposes. First, candidates were randomly assigned to triplets, so the gender composition of each group is orthogonal to candidate quality. Second, each triplet was independently assessed by two HR managers, and the order in which managers encountered the five triplets was randomised. This means we can compare evaluations of the *same* candidates, in the *same* triplet, by two managers who encounter that triplet at different points in the sequence—one as her first assessment, another as her fifth. Any difference in how they evaluate female candidates within the triplet is therefore attributable to sequence position, not to candidate or group characteristics. Third, the full set of assessments took each manager approximately one hour, providing sufficient scope for fatigue to accumulate across rounds.

Table A.13 shows that there is a clear decline in female candidates' performance over the sequence of an evaluator's assessments. With every subsequent round, it becomes 3.4 percentage points less likely on average that a female candidate is ranked first in the triplet. In this setting we do not find evidence for any difference by manager gender, nor of a decrease in the quality of decisions. Table A.13 reports the ranking results excluding the firms whose managers participated in the current experiment, to ensure we rely on an entirely different sample of managers.

[1] We omit the firms whose managers partake in the control group in the main experiment in this paper for our main specification, and show the results are robust to including them.

Panel A: Treatment effects			
	(1) Avg. score	(2) Strict (>0.5 SD)	(3) Both experts
Information	0.096**	0.115**	0.057
	(0.041)	(0.044)	(0.047)
Peer-Justification	0.025	0.008	0.014
	(0.041)	(0.043)	(0.047)
Information & Peer-Justification	0.012	-0.002	-0.018
	(0.042)	(0.050)	(0.056)
Control mean	0.561	0.618	0.647
N	1,323	798	719
Pair FE	Yes	Yes	Yes

Panel B: Treatment effects by round			
	(1) Avg. score	(2) Strict (>0.5 SD)	(3) Both experts
Information	-0.035	-0.010	-0.063
	(0.055)	(0.064)	(0.069)
Peer-Justification	-0.055	-0.047	-0.031
	(0.055)	(0.068)	(0.069)
Information & Peer-Justification	-0.082	-0.086	-0.078
	(0.059)	(0.073)	(0.078)
Second half=1	-0.176***	-0.137**	-0.137*
	(0.058)	(0.068)	(0.072)
Information × Second half=1	0.258***	0.239***	0.224**
	(0.077)	(0.085)	(0.097)
Peer-Justification × Second half=1	0.155**	0.103	0.082
	(0.074)	(0.092)	(0.095)
Information & Peer-Justification × Second half=1	0.182**	0.160	0.109
	(0.082)	(0.100)	(0.104)
Control mean	0.561	0.618	0.647
N	1,323	798	719
Pair FE	Yes	Yes	Yes

Notes: Column (1) restricts to pairs where the female candidate has a higher average expert score. Column (2) requires the score difference to exceed 0.5 standard deviations. Column (3) restricts to pairs where both experts favour the female candidate. Panel B additionally interacts treatment indicators with a late-round indicator (rounds 7–12). Standard errors are clustered at the judge level and all regressions include pair fixed effects. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table A.12: Sensitivity of treatment effects to expert favourite definition

	All		Male Manager		Female Favourite	
	(1) By Round	(2) Linear	(3) By Round	(4) Linear	(5) By Round	(6) Linear
Round 2 × Female	0.008		-0.014		-0.111	
	(0.062)		(0.110)		(0.093)	
Round 3 × Female	-0.100*		-0.220*		-0.233**	
	(0.059)		(0.113)		(0.094)	
Round 4 × Female	-0.027		-0.122		-0.071	
	(0.061)		(0.111)		(0.092)	
Round 5 × Female	-0.152**		-0.256**		-0.283***	
	(0.061)		(0.112)		(0.095)	
Round × Female		-0.034***		-0.062***		-0.051**
		(0.013)		(0.023)		(0.020)
Female × Male Mgr			0.013	-0.006		
			(0.101)	(0.097)		
Round 2 × Female × Male Mgr			0.036			
			(0.136)			
Round 3 × Female × Male Mgr			0.168			
			(0.140)			
Round 4 × Female × Male Mgr			0.107			
			(0.141)			
Round 5 × Female × Male Mgr			0.146			
			(0.136)			
Female × Male Mgr × Round				0.036		
				(0.029)		
Female × Fem. Favourite					-0.180**	-0.118
					(0.088)	(0.090)
Round 2 × Female × Fem. Favourite					0.228*	
					(0.117)	
Round 3 × Female × Fem. Favourite					0.229*	
					(0.120)	
Round 4 × Female × Fem. Favourite					0.102	
					(0.117)	
Round 5 × Female × Fem. Favourite					0.214*	
					(0.120)	
Female × Fem. Favourite × Round						0.030
						(0.026)
N	15,506	15,506	13,408	13,408	14,808	14,808
R^2	0.224	0.223	0.243	0.243	0.229	0.228
Firm × Vignette × Sector FE	Yes	Yes	Yes	Yes	Yes	Yes
Candidate FE	Yes	Yes	Yes	Yes	Yes	Yes
Mean dep. var.	0.342	0.342	0.341	0.341	0.342	0.342

Notes: OLS estimates. The dependent variable is an indicator equal to one if the candidate is ranked first among three candidates in the vignette. *Female* is an indicator for female candidates. *Round* indicates the presentation order of the vignette (1–5). Columns (1)–(2) pool all managers; columns (3)–(4) interact with an indicator for male managers; columns (5)–(6) interact with an indicator for whether the female candidate has the highest leave-one-out expert score in the group. Odd-numbered columns include round as a factor variable with Round 1 as the omitted category; even-numbered columns model round linearly. All specifications include firm × vignette × sector and candidate fixed effects. Standard errors clustered at the firm level in parentheses. The sample excludes firms whose managers participated in the committee experiment control group. Excluding these managers does not substantively affect the results. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A.13: Effect of candidate gender on ranking outcomes (excl. committees control group)

F Machine Learning Procedures

This section describes the machine-learning procedures used to examine treatment effect heterogeneity.

F.1 Variables and variable construction

Each observation is a judge-level assessment of a candidate pair. The outcome is `womanWins`, an indicator equal to one when the judge selects the female candidate. Treatments are coded as in the main analysis: 1 is control, 2 is Information, 3 is Peer-Justification, and 4 is the Information & Peer-Justification treatment. Early rounds are rounds 1–6 (`roundJudge` ≤ 6); late rounds are rounds 7–12.

The covariate matrix combines several families of predictors, summarised in Table A.14.

Family	Representative variables and role
Expert-evaluation variables	`overallExp`, `ConceptExp`, `MarketExp`, `GrowthExp`, `FinanceExp`, `PlanExp`, `SenseExp`, and `PresentExp`. These variables summarize expert assessments of candidate quality and presentation.
Text and language variables	`valenceMean`, `engagementMean`, `agentic_per_1000`, `communal_per_1000`, `genbit_avg_bias`, and the `empath_*` measures. These variables capture sentiment, linguistic content, and gendered language in submissions.
Presentation and content variables	`appearanceAppropriateExp`, `confidenceExp`, `arrogantExp`, `certainExp`, `mentionsIdea`, `mentionsMarket`, `dressedFormalEnu`, `careOutfitEnu`, `generalAppearance`, `confidenceEnu`, `teamworkEnu`, `Articulate`, and `specificExamples`. These variables measure visible presentation style, business content, and other candidate characteristics.

Table A.14: Variables used in the machine-learning procedures

Candidate-level covariates—expert evaluation sub-scores, text and language features, presentation assessments, business content indicators, and candidate age—are encoded as *standardised female-minus-male differences* within each pair. A positive value indicates that the female candidate scores higher on that dimension. All continuous covariates are standardised to mean zero and unit variance before estimation.

F.2 Abadie-style judge-level heterogeneity procedure

We use the endogenous stratification approach of Abadie, Chingos, and West (2018) to examine heterogeneity in treatment effects by predicted baseline performance. The key idea is to construct a prognostic score—the predicted early-round probability that a female candidate wins—and then estimate grouped average treatment effects (GATEs) for observations with low versus high predicted performance. Our implementation exploits the temporal structure of the experiment: because the main reduced-form results show that treatment effects are concentrated in late rounds (rounds 7–12), while early rounds (rounds 1–6)

exhibit no detectable treatment effects, we use early-round data to train the prognostic model and apply it to the all-round, early-round, and late-round samples.

Training data and justification. Within each repetition, the prognostic model is trained on the auxiliary half of the early-round candidate pairs, pooling observations across all four treatment arms. The full early-round sample contains 1,431 judge–pair observations, of which 327 are in the control group. Pooling treatment arms therefore provides a larger training sample than using the control group alone. This choice is justified by the null treatment effects in early rounds: given the absence of detectable treatment effects in rounds 1–6 (see Table 3, Panel B), we treat all early-round data as pre-treatment with respect to the fatigue mechanism that generates late-round bias. The early-round null also serves as an internal validation: when the trained model is applied to the held-out early-round sample, it finds no treatment effect heterogeneity, confirming that the prognostic score does not generate spurious patterns where none exist.

Covariate selection. The prognostic model excludes the overall expert evaluation score (`overallExp`), which is the single strongest predictor of `womanWins`. Including it would cause the prognostic grouping to collapse to the expert-favoured subgroup analysis already reported in the reduced-form results. The model retains the expert sub-scores (concept, market potential, growth, finance, plan quality, sense, and presentation), expert assessments of presentation style (appearance, confidence, arrogance, certainty), as well as candidate age, text and language features, enumerator-coded measures, and business content indicators. This covariate set captures a richer, multi-dimensional profile of candidate quality—including specific dimensions of the business proposal and presentation—while avoiding the single summary score that drives the reduced-form heterogeneity results. Because all candidate-level covariates are female-minus-male differences (see Section F.1), the prognostic score predicts relative female success within each pair.

Sample splitting. Within each repetition, the procedure draws a random 50/50 split of early-round candidate pairs. Even though the outcome is at the judge level, the split is performed over unique pair identifiers to ensure that all observations from a given candidate pair are assigned to the same subsample. We fit the model on one half and score the other, then reverse their roles, so every observation is scored by a model trained on different candidate pairs

Model estimation. On the auxiliary subsample, we estimate an elastic net for the binary outcome `womanWins` using `lassoglm`. The covariate matrix consists of columns 16 onward from the exported ML data file—expert sub-scores, age, text/language features, expert and enumerator presentation assessments, business content indicators, and interpersonal measures (see Table A.14)—excluding only the overall expert score in column 15. We perform a grid search over 25 values of the elastic-net mixing parameter $\alpha \in [0.01, 1]$ and 25 values of the regularisation parameter $\lambda \in [\exp(-10), \exp(-1)]$, using five-fold cross-validation with two Monte Carlo replications to evaluate deviance at each grid point. The (α, λ) pair minimising cross-validated deviance is selected. We then refit the model using only the covariates with non-zero coefficients from the initial regularised fit, estimating an unpenalised logistic regression on the selected variables. The procedure is therefore an elastic-net selection step followed by an unpenalised post-selection logistic regression.

Scoring and grouping. The fitted model is applied to three target samples, drawing only from pairs not used in training: (i) the complementary half of early-round pairs; (ii) the complementary half of the late-round pairs; and (iii) the combination of these two samples. Let $\hat{m}_p$ denote the predicted probability that the woman wins. Within each target sample, observations are divided at the median of $\hat{m}_p$ into low-

and high-predicted-performance groups.

Treatment effect estimation. Within each group g, the grouped average treatment effect for treatment arm t relative to the control group is:

$$\widehat{\tau}_{g,t} = \mathbb{E}[Y_{jp} \mid G_p = g, T_j = t] - \mathbb{E}[Y_{jp} \mid G_p = g, T_j = 1]\,.$$

where Y_{jp} is `womanWins`, $T_j = 1$ denotes the control group, and G_p is the predicted-performance group. Treatment-specific and control-group means within each group are reported alongside the GATEs.

Inference. We average estimates over 100 random pair splits. To calculate standard errors, we resample judges within treatment arms and repeat the full fitting, scoring, grouping, and treatment-effect procedure in 1,000 bootstrap replications. We report 90% confidence intervals based on the bootstrap standard errors

Interpretation. This procedure is an exploratory heterogeneity exercise rather than a formal structural estimator. It asks whether treatment effects differ systematically between observations that appear weaker or stronger according to a model trained in the pre-fatigue early rounds. The late-round estimates are the primary output, as they are fully out-of-sample relative to the training data. The early-round estimates serve as a placebo check.

F.3 Chernozhukov-style generic machine learning procedure

We complement the Abadie prognostic-score approach with the generic machine learning (GML) framework of Chernozhukov, Demirer, Duflo, and Fernández-Val (2025) to test for and characterise heterogeneity in the effect of Information messaging. Whereas the Abadie procedure stratifies by predicted *baseline* performance, the GML framework directly estimates pair-level conditional average treatment effects (CATEs) and tests whether they vary systematically across observations. The procedure yields three outputs: a best linear predictor (BLP) test for heterogeneity, grouped average treatment effects (GATES) that summarize treatment effects by predicted effect size, and classification analysis (CLAN) that describes the characteristics of the most- and least-affected groups.

Treatment comparison. Because Information messaging is the active treatment margin, we pool the two arms that include it (treatment arms 2 and 4, both with $\texttt{info} = 1$) as the treated group, compared against the control group (treatment arm 1).

Estimation sample. The sample is restricted to late rounds and treatment arms 1, 2, and 4, yielding 1,006 judge–pair observations, 238 candidate pairs, and 173 judges.

Sample splitting. Within each repetition, unique candidate pairs are randomly partitioned into auxiliary and main halves. All observations from a given candidate pair are assigned to the same subsample. The auxiliary half is used to train separate outcome models for the control and pooled treated groups; the main half is used for treatment effect estimation. This pair-level split ensures that information from a given candidate pair does not appear in both the training and estimation samples.

Model estimation. On the auxiliary subsample, we estimate separate elastic-net logistic regressions for the control arm and the pooled treated arm, modelling

$$\Pr(Y_{jp} = 1 \mid X_p, D_{jp} = d), \qquad d \in \{0, 1\},$$

with an ℓ_1/ℓ_2 penalty. The covariate vector X_p contains 35 candidate-pair characteristics: columns 15–28 and 35–55 of the exported machine-learning dataset. These include the overall expert score and exclude

the six Empath measures. We search over $\alpha \in \{0.05, 0.25, 0.50, 0.75, 1\}$ and 16 logarithmically spaced values of λ between $10^{-4.5}$ and $10^{-0.5}$. Tuning uses three-fold cross-validation, with all observations associated with a candidate pair assigned to the same fold. The final prediction rule remains penalised; no unpenalised post-selection refit is performed.

Predicted treatment effects. The fitted models are applied to the main subsample to predict outcomes under the pooled treated and control conditions. For candidate pair p,

$$S_p = \widehat{m}(X_p, D = 1) - \widehat{m}(X_p, D = 0).$$

Because the covariates are pair-level, S_p is constant across judge-level observations for a pair and is used as a ranking score.

BLP test for heterogeneity. Following Chernozhukov et al. (2025), we estimate the following BLP with candidate-pair fixed effects:

$$Y_{jp} = \mu_p + \beta_1(D_j - \pi) + \beta_2(D_j - \pi)(S_p - \bar{S}) + \varepsilon_{jp},$$

where μ_p is a candidate-pair fixed effect, D_j is the treatment indicator for judge j, and $\pi = 2/3$ is the design-implied treatment probability (two pooled arms versus one control arm). The coefficient β_1 captures the treatment effect at the mean predicted score; β_2 captures whether predicted heterogeneity translates into actual heterogeneity. A significant β_2 rejects homogeneous effects. Standard errors are clustered at the judge level.

GATES. Candidate pairs are ranked by S_p and divided into groups that are as nearly equal in size as possible. The primary specification uses two groups and the robustness specification uses three. Within each group, treatment effects are estimated using group-specific treatment indicators and candidate-pair fixed effects. Group labels follow the predicted-score ranking; estimates are not reordered ex post and monotonicity is not imposed.

CLAN. For each covariate, we estimate its mean within each GATES group. Comparing the characteristics of the most-affected group (highest predicted treatment effect) with the least-affected group reveals which observable features distinguish observations that benefit most from the treatment.

Inference. The procedure is repeated 500 times with deterministic pair-level splits generated from distinct seeds. For each split, 90% confidence intervals are constructed as $\hat{\theta} \pm 1.645 \times \hat{\sigma}$, where $\hat{\sigma}$ is the clustered standard error from the split-specific regression. The reported point estimate is the median of $\hat{\theta}$ across splits; the reported confidence bounds are the medians of the split-specific lower and upper bounds, following the variational inference approach in Chernozhukov et al. (2025).

F.4 Machine learning results

F.4.1 Abadie-style heterogeneity results

Table 3 in the main paper presents the Abadie-style grouped treatment effects. The prognostic model is trained on the auxiliary half of the early-round pairs, pooling all treatment arms and applied out-of-sample to late rounds (Panel C), the complementary half of early rounds (Panel B), and all rounds combined (Panel A). The model partitions observations at the median of the predicted probability that the female candidate wins.

F.4.2 Chernozhukov-style heterogeneity results

The estimated treatment effect at the mean predicted score is 8.4 percentage points (90% CI $[0.5, 16.6]$, $p = 0.082$). The BLP heterogeneity loading is positive but imprecisely estimated: $\widehat{\beta}_2 = 0.338$ (90% CI $[-0.454, 1.154]$, $p = 0.492$). We therefore do not reject homogeneous treatment effects. In the two-group specification, the estimated treatment effect is 5.0 percentage points in the below-median score group (90% CI $[-5.9, 16.0]$) and 12.1 percentage points in the above-median group (90% CI $[0.4, 23.7]$). The direct high-minus-low contrast is 7.5 percentage points (90% CI $[-8.5, 23.1]$, $p = 0.439$). The CLAN analysis reported in Table A.15 shows that the highest-score group contains candidate pairs in which the woman has stronger expert-assessed business and presentation characteristics, mentions operations more often, and is rated as more articulate. The woman also receives a higher expert rating for appropriate attire in this group.

F.4.3 Tercile-split results

As a robustness check, we divide candidate pairs into three groups based on the predicted treatment-effect score. We do not report the detailed results here, but note that the GATES point estimates increase from 4.4 percentage points in the lowest-score group to 9.2 percentage points in the middle group and 12.6 percentage points in the highest-score group. The direct G3–G1 contrast is 9.0 percentage points (90% CI $[-10.5, 28.1]$, $p = 0.447$). The BLP results are (mechanically) unchanged from the binary split, and the CLAN results support a broadly similar narrative.

	G1	G2	Diff.	p-value
Overall score (expert)	-0.48	0.52	1.01***	0.000
Concept (expert)	-0.36	0.34	0.72***	0.000
Market (expert)	-0.40	0.39	0.81***	0.000
Growth (expert)	-0.23	0.30	0.54***	0.002
Finance (expert)	-0.21	0.24	0.47***	0.007
Plan (expert)	-0.35	0.36	0.73***	0.000
Business sense (expert)	-0.39	0.45	0.86***	0.000
Presentation (expert)	-0.39	0.41	0.83***	0.000
Age	0.03	-0.06	-0.09	0.644
Valence (text)	0.08	-0.12	-0.19	0.305
Engagement (text)	0.05	-0.04	-0.10	0.574
Agentic language	-0.06	0.04	0.09	0.629
Communal language	-0.01	0.01	0.01	0.940
Gender bias (text)	0.03	-0.02	-0.05	0.762
Appropriate attire (expert)	-0.24	0.20	0.44**	0.010
Confidence (expert)	-0.40	0.46	0.90***	0.000
Arrogance (expert)	-0.02	0.10	0.12	0.525
Certainty (expert)	-0.43	0.43	0.87***	0.000
Confident, woman only (expert)	-0.38	0.33	0.75***	0.000
Mentions idea	-0.17	0.06	0.23	0.174
Mentions market	-0.12	0.08	0.19	0.269
Mentions competition	-0.02	0.01	0.03	0.878
Mentions operations	-0.39	0.44	0.83***	0.000
Mentions cost	-0.05	-0.01	0.06	0.745
Dressed formally (enum.)	-0.05	0.01	0.07	0.683
Care with outfit (enum.)	-0.08	0.06	0.17	0.379
General appearance (enum.)	-0.03	-0.00	0.05	0.797
Confidence (enum.)	-0.22	0.24	0.47***	0.010
Arrogance (enum.)	0.10	-0.12	-0.17	0.339
Teamwork (enum.)	-0.15	0.14	0.28	0.116
Mentions family (enum.)	-0.18	0.14	0.32*	0.081
Certainty (enum.)	-0.22	0.19	0.43**	0.018
Articulate	-0.29	0.29	0.60***	0.001
Specific examples	-0.11	0.21	0.32*	0.077
Gender bias, woman (text)	0.07	-0.09	-0.16	0.378

Notes: This table reports mean covariate values within the least-affected (G1) and most-affected (G2) GATES groups from the Chernozhukov et al. (2025) classification analysis (CLAN). All covariates are standardised differences between the female and male candidate in the pair, with positive values indicating that the female candidate scores higher. The treated group pools treatment arms 2 (Information) and 4 (Information & Peer-Justification); the control group is treatment arm 1. The estimation sample is restricted to late rounds (rounds 7–12). Point estimates are medians across 500 random sample splits.

Table A.15: CLAN: Characteristics of most and least affected groups

G Treatment Effect Heterogeneity by Judge Gender

This section examines whether the main treatment effects differ by the gender of the evaluating judge. We estimate a pooled specification allowing treatment effects to vary by judge gender:

$$\begin{aligned} Y_{jcp} = \alpha &+ \sum_k \beta_k \,\text{Treatment}_k + \gamma \,\text{Late round} + \eta \,\text{Female}_j \\ &+ \sum_k \delta_k (\text{Treatment}_k \times \text{Late round}) + \sum_k \phi_k (\text{Treatment}_k \times \text{Female}_j) \\ &+ \rho (\text{Late round} \times \text{Female}_j) \\ &+ \sum_k \psi_k (\text{Treatment}_k \times \text{Late round} \times \text{Female}_j) + \mu_p + \varepsilon_{jcp}. \end{aligned} \tag{G.13}$$

where j indexes judges, c committees, and p candidate pairs; μ_p denotes pair fixed effects. We cluster standard errors at the judge level.

Table A.16 presents results using the binary second-half indicator (rounds 7–12). The fatigue-induced decline in female candidate selection and its prevention by the Information treatment are concentrated among male judges. For male judges, the Information treatment effect in late rounds is large and significant ($p = 0.004$ for female wins, $p < 0.001$ for expert-favoured female wins). Peer-Justification also increases expert-favoured female wins among male judges in late rounds relative to the control group ($p = 0.043$) Female judges show no significant treatment effects in either early or late rounds.

	(1) Woman Wins	(2) Expert-favoured Woman Wins	(3) Expert-favoured Man Wins
Information	-0.049	-0.033	0.007
	(0.044)	(0.029)	(0.032)
Peer-Justification	-0.000	-0.011	-0.019
	(0.044)	(0.028)	(0.035)
Information & Peer-Justification	-0.043	-0.047	-0.007
	(0.047)	(0.030)	(0.035)
Late Round	-0.118**	-0.097***	0.026
	(0.050)	(0.029)	(0.036)
Information × Late Round	0.177***	0.160***	-0.024
	(0.062)	(0.040)	(0.043)
Peer-Justification × Late Round	0.065	0.069*	-0.003
	(0.059)	(0.038)	(0.042)
Information & Peer-Justification × Late Round	0.108*	0.106**	-0.007
	(0.062)	(0.043)	(0.046)
Female judge	-0.006	0.004	-0.012
	(0.065)	(0.045)	(0.042)
Information × Female	0.086	0.085	0.007
	(0.098)	(0.058)	(0.066)
Peer-Justification × Female	-0.074	-0.048	0.056
	(0.086)	(0.060)	(0.061)
Information & Peer-Justification × Female	-0.002	0.042	0.041
	(0.082)	(0.057)	(0.057)
Late Round × Female	0.049	0.054	0.001
	(0.093)	(0.064)	(0.060)
Information × Late Round × Female	-0.191	-0.176*	0.018
	(0.132)	(0.092)	(0.086)
Peer-Justification × Late Round × Female	0.012	0.012	-0.001
	(0.113)	(0.083)	(0.074)
Information & Peer-Justification × Late Round × Female	-0.034	-0.081	-0.026
	(0.124)	(0.085)	(0.079)
Pair FE	Yes	Yes	Yes
Control mean (early, male)	0.504	0.268	0.311
p: Information late (male)	0.004	0.000	0.601
p: Peer-Justification late (male)	0.139	0.043	0.467
p: Information & Peer-Justification late (male)	0.170	0.058	0.691
p: Information late (female)	0.783	0.609	0.823
p: Peer-Justification late (female)	0.971	0.699	0.461
p: Information & Peer-Justification late (female)	0.751	0.722	0.983
N	2,858	2,858	2,858

Notes: OLS estimates of treatment effects on committee decision outcomes, interacted with evaluation timing and judge gender. The dependent variable in columns 1 is an indicator for the female candidate being selected; column 2 indicates whether an expert-favoured woman is selected; column 3 indicates whether an expert-favoured man is selected. Treatment indicators are each interacted with *Second half*, an indicator for rounds 7–12, and the judges' gender. "Control mean (early, male)" reports the dependent variable mean for men in the control group in rounds 1–6. All specifications include candidate-pair fixed effects. Standard errors clustered at the judge level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.16: Treatment effects by evaluation timing and judge gender (binary)

H Long-term labour market outcomes

Around 18 months following the competition, we collect follow-up data to assess what happened to the candidates after the competition. We collect data from 90 of the 100 candidates on their employment status, labour market outcomes, and some questions relating to their current business (plans). To understand whether judges recommended different types of candidates across treatments, we run the following regression pooling the treatments, where the observation is the labour market outcomes of individual i:

$$\begin{aligned} Y_i = &\alpha + \beta_1 \texttt{Score_Control}_i + \beta_2 \texttt{Score_Information}_i + \\ &\beta_3 \texttt{Score_Justification}_i + \beta_4 \texttt{Score_Both_treatments}_i + \\ &\zeta \texttt{Winner}_i + \varepsilon_i \end{aligned}$$

where `Score_X` is the number of pairwise comparisons a candidate won in treatment arm X, and `Winner` is an indicator for having won the prize. Table A.17 reports the results from this regression.[2] The outcomes Y include indicators for transitioning into self-employment, being self-employed, and being wage-employed; total income; total investment in the respondent's business since the competition; the number of steps taken to establish a business; and an indicator for having taken out a business loan since the competition. These measures capture participants' labour-market outcomes following the competition.

Table A.17: Long-term outcomes and competition performance

	Transition into self-employment	Self-employed	Wage-employed	Total income (ETB1000)	Total investment (ETB1000)	Total steps taken	Has loan
Score control	0.025	-0.003	0.006	0.029	0.036	0.233	-0.024
	(0.02)	(0.02)	(0.02)	(0.04)	(0.06)	(0.23)	(0.02)
Score Information	-0.002	0.016	-0.005	-0.004	0.060	0.291	0.029
	(0.02)	(0.02)	(0.02)	(0.05)	(0.07)	(0.27)	(0.02)
Score Peer-Justification	-0.038**	-0.042*	0.001	-0.018	-0.051	-0.060	0.003
	(0.02)	(0.02)	(0.03)	(0.05)	(0.07)	(0.31)	(0.02)
Score Information &	0.019	0.024	0.002	0.058	0.017	-0.161	-0.027
Peer-Justification	(0.02)	(0.03)	(0.03)	(0.06)	(0.09)	(0.32)	(0.02)
Winner	-0.020	0.097	0.125	0.057	-0.250	-0.783	0.173
	(0.13)	(0.20)	(0.15)	(0.40)	(0.67)	(2.23)	(0.21)
Sample mean	0.138	0.368	0.770	31.278	159.954	6.161	0.345
N	87	87	87	87	87	87	87

Notes: This table describes the relationship between labor market outcomes and candidate performance in each of the treatment arms. The transition into self-employment is a dummy equal to one if the respondent was not in self-employment before but is after the competition. Self-employed and wage-employed are dummies. Total income is monthly income out of any source (in ETB 1000). Total investment is the amount the respondent has invested in their business since the competition (in ETB 1000), estimated using Poisson regression. Total steps taken is the number of steps (from a fixed list) a respondent has taken starting a business. Has loan is an indicator for whether the respondent has a business loan. All monetary variables are winsorized at the 95th percentile. $^{*}\ p < 0.10$, $^{**}\ p < 0.05$, $^{***}\ p < 0.01$

[2] We exclude transitions out of self-employment as an outcome because few respondents report such transitions.

Table A.18: Long-term outcomes and competition performance by round period

	Transition into self-employment	Self-employed	Wage-employed	Total income (ETB 1000)	Total investment (ETB 1000)	Total steps taken	Has loan
Panel A: Early rounds							
Score control	0.008	-0.005	-0.000	0.001	-0.022	0.175	-0.012
	(0.01)	(0.01)	(0.01)	(0.03)	(0.04)	(0.16)	(0.01)
Score Information	-0.005	0.011	-0.004	0.022	0.052	0.225	0.021
	(0.01)	(0.02)	(0.01)	(0.03)	(0.04)	(0.17)	(0.02)
Score Peer-Justification	-0.015	-0.007	0.005	-0.008	0.014	0.004	0.006
	(0.01)	(0.02)	(0.02)	(0.04)	(0.06)	(0.19)	(0.02)
Score Information & Peer-Justification	-0.002	-0.005	-0.000	0.005	-0.038	-0.171	-0.020
	(0.01)	(0.01)	(0.01)	(0.04)	(0.05)	(0.16)	(0.01)
Winner	0.038	0.109	0.147	0.266	-0.078	-0.803	0.118
	(0.12)	(0.20)	(0.15)	(0.39)	(0.66)	(2.22)	(0.21)
Sample mean	0.133	0.367	0.756	30.412	156.589	6.189	0.344
N	90	90	90	90	90	90	90
Panel B: Late rounds							
Score control	0.009	-0.012	-0.010	0.018	0.047	-0.022	-0.021
	(0.01)	(0.02)	(0.02)	(0.03)	(0.06)	(0.20)	(0.02)
Score Information	-0.007	0.004	-0.004	-0.014	0.039	0.071	0.007
	(0.01)	(0.02)	(0.02)	(0.04)	(0.06)	(0.21)	(0.02)
Score Peer-Justification	-0.014	-0.027*	-0.003	0.001	-0.042	-0.011	-0.000
	(0.01)	(0.02)	(0.02)	(0.04)	(0.05)	(0.21)	(0.02)
Score Information & Peer-Justification	0.008	0.022	0.009	0.033	0.021	0.081	0.002
	(0.01)	(0.02)	(0.02)	(0.04)	(0.07)	(0.23)	(0.02)
Winner	0.004	0.113	0.167	0.192	-0.245	-0.556	0.131
	(0.14)	(0.20)	(0.13)	(0.37)	(0.63)	(2.20)	(0.20)
Sample mean	0.133	0.367	0.756	30.412	156.589	6.189	0.344
N	90	90	90	90	90	90	90
Fixed effects	No	No	No	No	No	No	No

Notes: This table describes the relationship between labor market outcomes and candidate performance in each of the treatment arms, separately for early rounds (1–6) and late rounds (7–12). The transition into self-employment is a dummy equal to one if the respondent was not in self-employment before but is after the competition. Self-employed and wage-employed are dummies. Total income is monthly income out of any source (in ETB 1000). Total investment is the amount the respondent has invested in their business since the competition (in ETB 1000), estimated using Poisson regression. Total steps taken is the number of steps (from a fixed list) a respondent has taken starting a business. Has loan is an indicator for whether the respondent has a business loan. All monetary variables are winsorized at the 95th percentile. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

The results, presented in Table A.17, show that overall, performance in the competition has limited

predictive power for long-term labor market outcomes. This finding aligns with previous studies, such as McKenzie and Sansone (2019), which suggest that accurately predicting the future success of aspiring entrepreneurs based on competition performance is inherently difficult.

Despite the limited predictive power across most outcomes, there are a few notable patterns. Candidates who performed well in the control treatment (i.e. received high scores in the absence of any organisational or social pressure) are (non-significantly) more likely to transition into self-employment. Interestingly, those who scored well amongst control group judges also tended to invest more in their businesses. In contrast, candidates who performed well amongst judges in the Peer-Justification treatment are less likely to transition into self-employment and to be in self-employment. Given the large number of regressions done here these results are unlikely to survive rigorous multiple-hypothesis testing.

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